



Anil Seth

Building a Web Application Using Mochiweb

Mochiweb is an Erlang library for building lightweight HTTP servers. Using Fedora 20, the author leads readers through the steps to building a simple Web application.

This is the era of the app. The sheer number of them on various mobile devices is mind boggling. An app has to look good on the mobile device, but what about the backend? Will it scale if the app succeeds? How much effort should one put in to ensure scalability upfront, e.g., by using as many cores and processors available? How hard would it be to write a Web service in Erlang?

In this article, let's explore the Mochiweb framework by creating a simple application in which the Web application gets a keyword and responds with a list of values.

The template engine we will use is EryDTL, which is an Erlang implementation of Django Template Language.

Getting started

Let's experiment using Fedora 20. The steps are as follows:

- Install the *erlang-mochiweb*, *erlang-erydtl* and *erlang-rebar* packages
- Modify *mochiwebapp.template* in */usr/lib64/erlang/lib/mochiweb-2.4.2/support/templates/* to remove references to files not in the distribution—*Makefile*, *.gitignore* and *rebar*. The updated template will appear as follows:

```
%% *- erlang *- (variables, [{appid, "mochiwebapp"},
    {author, "Mochi Media <dev@mochimedia.com>"},
    {year, "2010"},
    {version, "0.1"},
    {port, 8080},
    {dest, "{{appid}}"}].
{dir, "{{dest}}".
{template, "mochiwebapp_skel/src/mochiapp.app.src",
  "{{dest}}/src/{{appid}}.app.src".
{template, "mochiwebapp_skel/src/mochiapp.erl", "{{dest}}/
src/{{appid}}.erl".
{template, "mochiwebapp_skel/src/mochiapp_app.erl",
  "{{dest}}/src/{{appid}}_app.erl".
{template, "mochiwebapp_skel/src/mochiapp_deps.erl",
  "{{dest}}/src/{{appid}}_deps.erl".
{template, "mochiwebapp_skel/src/mochiapp_sup.erl",
  "{{dest}}/src/{{appid}}_sup.erl".
{template, "mochiwebapp_skel/src/mochiapp_web.erl",
```

```
"{{dest}}/src/{{appid}}_web.erl".
{template, "mochiwebapp_skel/start-dev.sh", "{{dest}}/start-
dev.sh".
{template, "mochiwebapp_skel/priv/www/index.html", "{{dest}}/
priv/www/index.html".
{file, "mochiwebapp_skel/rebar.config", "{{dest}}/rebar.config".
{chmod, 8755, "{{dest}}/start-dev.sh".
```

Create a minimal app as follows:

```
$ cd /usr/lib64/erlang/lib/mochiweb-2.4.2/support/
$ rebar create template=mochiwebapp dest=~/.mochiwebx
appid=mochiwebx
```

Test it:

```
$ cd ~/.mochiwebx
$. /start-dev.sh
```

Open *http://localhost:8080* in a Web browser and you should see *mochiwebx* running.

Creating the template

Create the directory templates in *~/.mochiwebx*. Create the template file, *templates/webapp.dtl* with the following content:

```
<html>
<body>
  <form method="GET">
    <p>Key: <input type="text" name="key"></p>
    <input type="submit">
  </form>
  {% for key,value in the_list %}
    <p> {{ key }} Value {{ value }} </p>
  {% endfor %}
</body>
</html>
```

We iterate over the elements of a list and display the values of each element in a row.